ERA 5 calibration to Elexon power

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Bias

Low wind events

- Important at aggregate GB level for system operation and security of supply.
- We define low wind events as periods of time when the total GB wind power generation is below a certain threshold (e.g., 5% of installed capacity).
- We can analyze the distribution of these events in terms of their duration, frequency, and severity (e.g., cumulative shortfall).
- We can also compare the observed low wind events with those simulated by the ERA5 reanalysis data to assess the accuracy of the reanalysis in capturing these events.
- Ideally we'll model shortfall events, considering demand and other generation sources, but for now we focus on wind generation only.

Capacity Factor distribution

